

Name: Gaurika Mittal

Sap ID: 590034892

Batch: 15

Experiment-20

1. Algorithm

1. Start.
2. Declare integer variables m , n , $choice$, i , and j .
3. Input two positive integers m and n .
4. Check whether m and n are positive and $m \leq n$.
5. If the input is invalid, display an error message and ask for m and n again.
6. Display the menu:
 1. Multiplication Table
 2. Addition Table
 3. Subtraction Table
 4. Division Table
 5. Exit
7. Input the user's choice.
8. Use a `switch` statement to perform the selected operation.
9. If `choice = 1`:
 - For every integer from m to n , print its first 10 multiples.
10. If `choice = 2`:
 - For every integer from m to n , print its addition table using numbers 1 to 10.
11. If `choice = 3`:
 - For every integer from m to n , print its subtraction table using numbers 1 to 10 as subtrahends.
12. If `choice = 4`:
 - For every integer from m to n , divide it by numbers 1 to 10.
 - Display each result up to two decimal places.
13. If `choice = 5`, display an exit message and terminate the program.
14. For any other choice, display `Invalid menu choice`.
15. Repeat the menu until the user selects option 5.
16. Stop.

2. Test-Case Table

Test Case	Input	Choice	Expected Output	Results
1	$m = 2, n = 3$	1	First 10 multiples of 2 and 3	Valid

2	m = 2, n = 3	2	Addition tables of 2 and 3 from 1 to 10	Valid
3	m = 2, n = 3	3	Subtraction tables of 2 and 3 using 1 to 10	Valid
4	m = 2, n = 3	4	Division tables of 2 and 3 by 1 to 10, rounded to 2 decimal places	Valid
5	m = 1, n = 1	1	First 10 multiples of 1	Boundary
6	m = 5, n = 2	-	Invalid input. Enter positive integers with m <= n.	Invalid
7	m = -2, n = 5	-	Invalid input. Enter positive integers with m <= n.	Invalid
8	m = 2, n = 3	7	Invalid menu choice.	Invalid
9	m = 2, n = 3	5	Program exits	Valid

3. C Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int m, n, choice, i, j;
```

```
    do
```

```
    {
```

```
        printf("Enter two positive integers m and n (m <= n): ");
```

```
        scanf("%d %d", &m, &n);
```

```
        if (m <= 0 || n <= 0 || m > n)
```

```
        {
```

```
            printf("Invalid input. Enter positive integers with m <= n.\n");
```

```
        }
```

```
    } while (m <= 0 || n <= 0 || m > n);
```

```
    do
```

```
    {
```

```
        printf("\nMENU\n");
```

```
        printf("1. Print Multiplication Tables\n");
```

```
        printf("2. Print Addition Tables\n");
```

```
        printf("3. Print Subtraction Tables\n");
```

```

printf("4. Print Division Tables\n");
printf("5. Exit\n");

printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice)
{
    case 1:
        for (i = m; i <= n; i++)
        {
            printf("\nMultiplication Table of %d\n", i);

            for (j = 1; j <= 10; j++)
            {
                printf("%d x %d = %d\n", i, j, i * j);
            }
        }
        break;

    case 2:
        for (i = m; i <= n; i++)
        {
            printf("\nAddition Table of %d\n", i);

            for (j = 1; j <= 10; j++)
            {
                printf("%d + %d = %d\n", i, j, i + j);
            }
        }
        break;

    case 3:
        for (i = m; i <= n; i++)
        {
            printf("\nSubtraction Table of %d\n", i);

            for (j = 1; j <= 10; j++)
            {
                printf("%d - %d = %d\n", i, j, i - j);
            }
        }
        break;

    case 4:
        for (i = m; i <= n; i++)
        {
            printf("\nDivision Table of %d\n", i);

            for (j = 1; j <= 10; j++)
            {
                printf("%d / %d = %.2f\n",
                    i, j, (float)i / j);
            }
        }
}

```

```

        break;

    case 5:
        printf("Exiting program...\n");
        break;

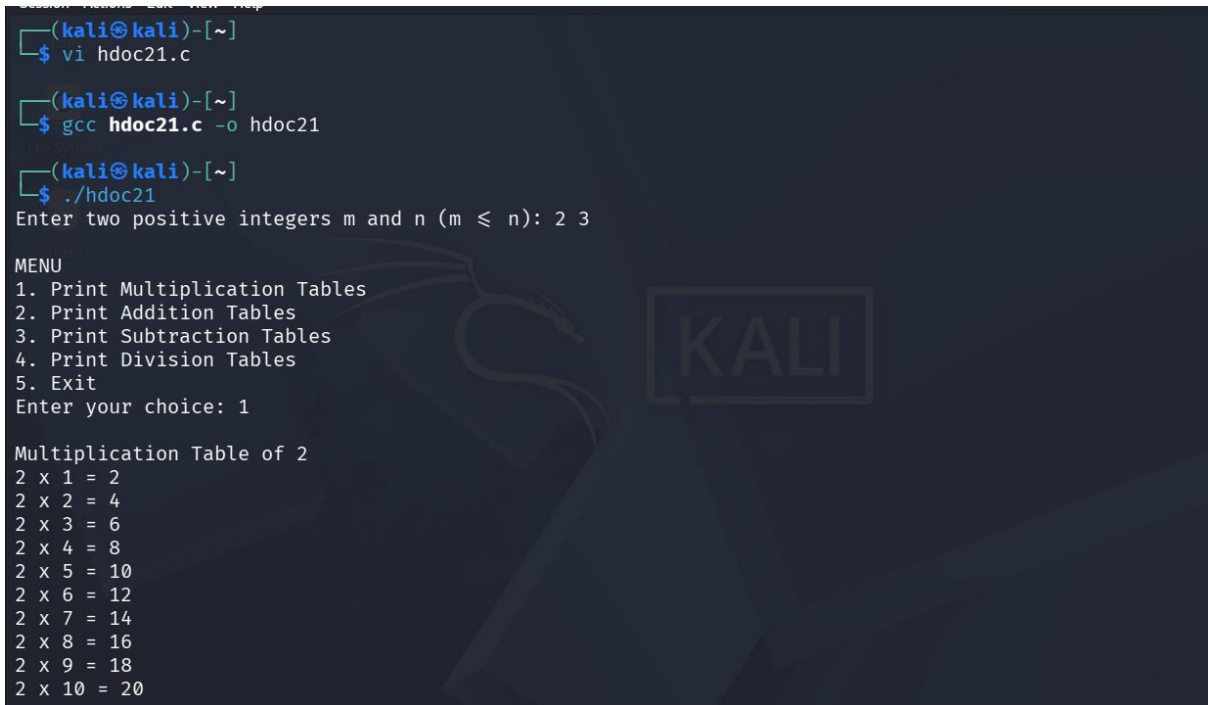
    default:
        printf("Invalid menu choice.\n");
    }

} while (choice != 5);

return 0;
}

```

4. Compilation and Execution



```

(kali㉿kali)-[~]
$ vi hdoc21.c

(kali㉿kali)-[~]
$ gcc hdoc21.c -o hdoc21

(kali㉿kali)-[~]
$ ./hdoc21
Enter two positive integers m and n (m ≤ n): 2 3

MENU
1. Print Multiplication Tables
2. Print Addition Tables
3. Print Subtraction Tables
4. Print Division Tables
5. Exit
Enter your choice: 1

Multiplication Table of 2
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20

```

Multiplication Table of 3

3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30

MENU

1. Print Multiplication Tables
2. Print Addition Tables
3. Print Subtraction Tables
4. Print Division Tables
5. Exit

Enter your choice: 2

Addition Table of 2

2 + 1 = 3
2 + 2 = 4
2 + 3 = 5
2 + 4 = 6
2 + 5 = 7
2 + 6 = 8
2 + 7 = 9
2 + 8 = 10

2 + 8 = 10
2 + 9 = 11
2 + 10 = 12

Addition Table of 3

3 + 1 = 4
3 + 2 = 5
3 + 3 = 6
3 + 4 = 7
3 + 5 = 8
3 + 6 = 9
3 + 7 = 10
3 + 8 = 11
3 + 9 = 12
3 + 10 = 13

MENU

1. Print Multiplication Tables
2. Print Addition Tables
3. Print Subtraction Tables
4. Print Division Tables
5. Exit

Enter your choice: 3

Subtraction Table of 2

2 - 1 = 1
2 - 2 = 0
2 - 3 = -1
2 - 4 = -2

```
2 - 5 = -3
2 - 6 = -4
2 - 7 = -5
2 - 8 = -6
2 - 9 = -7
2 - 10 = -8
```

Subtraction Table of 3

```
3 - 1 = 2
3 - 2 = 1
3 - 3 = 0
3 - 4 = -1
3 - 5 = -2
3 - 6 = -3
3 - 7 = -4
3 - 8 = -5
3 - 9 = -6
3 - 10 = -7
```

MENU

1. Print Multiplication Tables
2. Print Addition Tables
3. Print Subtraction Tables
4. Print Division Tables
5. Exit

Enter your choice: 4

Division Table of 2

```
2 / 1 = 2.00
```

```
2 / 2 = 1.00
2 / 3 = 0.67
2 / 4 = 0.50
2 / 5 = 0.40
2 / 6 = 0.33
2 / 7 = 0.29
2 / 8 = 0.25
2 / 9 = 0.22
2 / 10 = 0.20
```

Division Table of 3

```
3 / 1 = 3.00
3 / 2 = 1.50
3 / 3 = 1.00
3 / 4 = 0.75
3 / 5 = 0.60
3 / 6 = 0.50
3 / 7 = 0.43
3 / 8 = 0.38
3 / 9 = 0.33
3 / 10 = 0.30
```

MENU

1. Print Multiplication Tables
2. Print Addition Tables
3. Print Subtraction Tables
4. Print Division Tables
5. Exit

Enter two positive integers m and n ($m \leq n$): 1 1

MENU

1. Print Multiplication Tables
2. Print Addition Tables
3. Print Subtraction Tables
4. Print Division Tables
5. Exit

Enter your choice: 1

Multiplication Table of 1

1 x 1 = 1
1 x 2 = 2
1 x 3 = 3
1 x 4 = 4
1 x 5 = 5
1 x 6 = 6
1 x 7 = 7
1 x 8 = 8
1 x 9 = 9
1 x 10 = 10

MENU

1. Print Multiplication Tables
2. Print Addition Tables
3. Print Subtraction Tables
4. Print Division Tables
5. Exit

Enter two positive integers m and n ($m \leq n$): 5 2
Invalid input. Enter positive integers with $m \leq n$.
Enter two positive integers m and n ($m \leq n$): -2 5
Invalid input. Enter positive integers with $m \leq n$.
Enter two positive integers m and n ($m \leq n$): 2 3

MENU

1. Print Multiplication Tables
2. Print Addition Tables
3. Print Subtraction Tables
4. Print Division Tables
5. Exit

Enter your choice: 7

Invalid menu choice.

MENU

1. Print Multiplication Tables
2. Print Addition Tables
3. Print Subtraction Tables
4. Print Division Tables
5. Exit

Enter your choice: 5

Exiting program...